✅ FULL DEPLOYMENT PROCESS (WITH GIT BRANCHES + REMOTES)

For Backend + Frontend on Linux Server

📌 0. KNOW YOUR SERVER PATHS

Backend Project

```
/home/bcladmin/birla-scm-node
```

Frontend Project

```
/home/bcladmin/birla-scm-react-v2
```

🚀 1. BACKEND DEPLOYMENT (birla-scm-node)

Step 1 → Move to backend folder

```
cd ~/birla-scm-node
```

Step 2 → Check local branches

```
git branch
```

You saw:

```
  airiddha-server
  dev-portal
  login
* main
  user-specific-request
```

Step 3 → Check remote branches

```
git branch -r
```

You saw:

```
origin/HEAD -> origin/login
origin/airiddha-server
origin/api-calls-reduce
origin/bcldpportal
origin/dev-portal
origin/development
origin/login
origin/main
origin/prod-bclportal
origin/prod-dpnportal
origin/prod-rccportal
origin/rfc
origin/user-specific-request
```

✔ Step 4 → Check remote URL

```
git remote -v
```

You saw:

```
origin  https://github.com/Subu1987/BirlaDepoPortalBackend.git (fetch)
origin  https://github.com/Subu1987/BirlaDepoPortalBackend.git (push)
```

So backend is connected correctly.

Step 5 → Checkout branch you want to deploy

Most common:

```
git checkout main
```

If deploying a prod branch:

```
git checkout prod-bclportal
git checkout prod-dpnportal
git checkout prod-rccportal
```

Step 6 → Pull latest code

```
git pull
```

If pulling a specific remote branch:

```
git pull origin prod-bclportal
```

Step 7 → Restart backend with PM2

Check PM2 services:

```
pm2 ls
```

You saw:

```
id 0 → server → online
```

Reload/Restart the backend:

```
pm2 reload 0
```

or

```
pm2 restart server
```

Check logs:

```
pm2 logs server
```

🌐 2. FRONTEND DEPLOYMENT (birla-scm-react-v2)

Step 1 → Move to frontend folder

```
cd ~/birla-scm-react-v2
```

Step 2 → Check local branches

```
git branch
```

You saw:

```
  api-calls-reducing
  delivery-dropdown
  dms-damage-data
* main
  master
  release/development
  so-create-change
```

Step 3 → Check remote branches

```
git branch -r
```

You saw:

```
origin/airiddha
origin/api-calls-reducing
origin/cnf-user-logic
origin/delivery-dropdown
origin/dms-damage-data
origin/kiran
origin/main
origin/master
origin/prod-bclportal
origin/prod-dpnportal
origin/prod-rccportal
origin/release/development
origin/so-create-change
```

✔ Step 4 → Check remote URL

```
git remote -v
```

You saw:

```
origin  https://github.com/Subu1987/BirlaDepoPortalFrontend.git (fetch)
origin  https://github.com/Subu1987/BirlaDepoPortalFrontend.git (push)
```

✔ Step 5 → Checkout branch to deploy

Example:

```
git checkout main
```

For production:

```
git checkout prod-bclportal
git checkout prod-dpnportal
git checkout prod-rccportal
```

✔ Step 6 → Pull latest frontend code

```
git pull
```

✔ Step 7 → Build the project

You are using `depl.sh`:

```
sudo sh depl.sh
```

OR manual build:

```
npm install
npm run build
```

Build output goes into:

```
build/
```

✔ Step 8 → Deploy build files to web folder

If your server uses Apache/Nginx:

```
sudo cp -r build/* /var/www/html/
```

OR if custom hosting:

```
sudo cp -r build/* ~/birla-scm-react/build/
```

🔄 3. ROLLBACK PROCESS

If something breaks, go to the last stable branch.

Backend rollback

```
cd ~/birla-scm-node
git checkout main   # or any old working branch
git pull
pm2 restart server
```

Frontend rollback

```
cd ~/birla-scm-react-v2
git checkout main
git pull
npm run build
sudo cp -r build/* /var/www/html/
```

🛡️ 4. CHECK SUDO ACCESS

Your command:

```
sudo -v
```

returns nothing = **you already have valid sudo privileges**.

🎯 5. QUICK SUMMARY (Production Deployment)

BACKEND

```
cd ~/birla-scm-node
git checkout prod-bclportal
git pull
pm2 reload server
```

FRONTEND

```
cd ~/birla-scm-react-v2
git checkout prod-bclportal
git pull
sudo sh depl.sh
sudo cp -r build/* /var/www/html/
```

📊 Deployment Comparison Table — Backend vs Frontend

Category	Backend (birla-scm-node)	Frontend (birla-scm-react-v2)
Purpose	Node.js API server	React UI Application
Main Folder	`/home/bcladmin/birla-scm-node`	`/home/bcladmin/birla-scm-react-v2`
Remote Repository	`BirlaDepoPortalBackend.git`	`BirlaDepoPortalFrontend.git`
Remote Branch Examples	`main`, `prod-bclportal`, `prod-dpnportal`, `prod-rccportal`	`main`, `prod-bclportal`, `prod-dpnportal`, `prod-rccportal`
Local Branch Examples	`main`, `airiddha-server`, `dev-portal`	`main`, `release/development`, `api-calls-reducing`
Command to Check Branch	`git branch`	`git branch`
Command to Check Remote Branches	`git branch -r`	`git branch -r`
Switch Branch	`git checkout <branch>`	`git checkout <branch>`
Pull Latest Code	`git pull`	`git pull`
Dependency Install	`npm install` (only if needed)	`npm install`
Build Command	❌ No build needed	`npm run build` or `sudo sh depl.sh`
Output Folder	No build folder	`build/` (auto-created)
Deployment Method	PM2 reload	Copy build files to web server
Start / Restart	`pm2 restart server` / `pm2 reload 0`	No PM2; static files served by Apache/Nginx
Check Logs	`pm2 logs server`	Browser console / server logs
Rollback Method	Checkout previous branch + restart PM2	Checkout previous branch + rebuild + copy files
Service State	Always running via PM2	Static build files served; no process needed
Verification	API health check (`curl`)	Browser check (`http://server-ip/`)

✔ Quick High-Level Summary

Backend

- Pull → Restart PM2
- No build
- Runs continuously

Frontend

- Pull → Build → Copy build files
- No PM2
- Served as static website